WORKED SOLUTION

STEP III 2010, Question 9

Mechanics • Motion on a Circle

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Read the question before you go on. It is in the STEP Support Programme's question database:

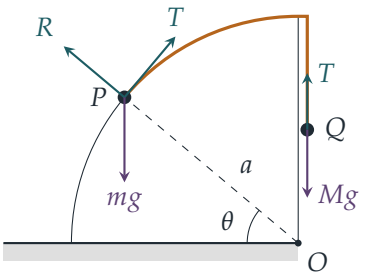
step.maths.org/questions/10-s3-q9

The key idea

Because the string is inextensible and the block is smooth, P and Q always move with the same speed, and the whole system conserves energy. That gives the speed of P as a function of θ with no differential equation to solve. Then P is moving on a circle, so resolving towards O gives the normal reaction. Staying in contact means that reaction stays non-negative, and a concavity argument shows that it's enough to check the top of the block.

Solution

The model. P and Q are particles, the string is light and inextensible, and the block is smooth, so the tension T is the same all along the string. Put P at angle θ on the arc. The string runs along the arc from P to the top of the block and then straight down beside the vertical face to Q.



Speed. When P has moved from $\theta = 0$ to θ it has travelled an arc of length $a\theta$, so Q has descended a distance $a\theta$. Both move with speed $v = a\dot{\theta}$. Meanwhile P has risen through a height $a\sin\theta$. The block is smooth and the string does no net work on the

system (it is inextensible), so energy is conserved. Starting from rest,

$$\tfrac{1}{2}(m+M)v^2 = Mga\theta - mga\sin\theta, \qquad \text{so} \qquad v^2 = \frac{2ga\,(M\theta - m\sin\theta)}{m+M}.$$

This is positive for $0 < \theta \le \frac{1}{2}\pi$, since $M\theta > m\theta \ge m\sin\theta$, so the system really does move off from rest.

Normal reaction. P moves on a circle of radius a about O, so its acceleration towards O is v^2/a. The forces on P are as follows:

- the reaction R along OP, pointing away from O;
- the tension T along the tangent, which has no radial component;
- the weight mg, whose component towards O is $mg\sin\theta$, since OP makes angle θ with the horizontal.

Resolving towards O,

$$mg\sin\theta - R = \frac{mv^2}{a} = \frac{2mg\,(M\theta - m\sin\theta)}{m+M}.$$

Hence

$$\begin{aligned} R &= \frac{mg\big[(m+M)\sin\theta - 2M\theta + 2m\sin\theta\big]}{m+M} \\ &= \boxed{\frac{mg\big[(3m+M)\sin\theta - 2M\theta\big]}{m+M}}. \end{aligned}$$

Ask yourself R must be non-negative for every θ in $[0, \frac{1}{2}\pi]$. Do we have to search for the minimum of R?

The condition. P stays in contact exactly when $R \ge 0$ throughout, that is, when

$$f(\theta) = (3m+M)\sin\theta - 2M\theta \ge 0 \quad \text{for } 0 \le \theta \le \tfrac{1}{2}\pi.$$

We don't need the minimum. Since $f''(\theta) = -(3m+M)\sin\theta \le 0$ on this interval, f is concave, so its graph lies on or above the chord joining its end points. We have $f(0) = 0$, so:

- *Necessity.* If P is still in contact at $\theta = \frac{1}{2}\pi$, then $f(\frac{1}{2}\pi) = 3m + M - \pi M \ge 0$.
- *Sufficiency.* If $f(\frac{1}{2}\pi) \ge 0$, then the chord from $(0,0)$ to $(\frac{1}{2}\pi, f(\frac{1}{2}\pi))$ lies on or above the axis. By concavity f lies on or above the chord, so $f \ge 0$ on the whole interval.

So the condition is $3m \ge (\pi - 1)M$, that is,

$$\frac{m}{M} \ge \frac{\pi - 1}{3},$$

as required. Note that $(\pi - 1)/3 \approx 0.714 < 1$, so this is consistent with $m < M$.

Checks

- At $\theta = 0$ we get $R = 0$. That's right: at the start P is at the foot of the block, the radius OP is horizontal, and P's weight is carried by the table.
- Resolving tangentially for P and vertically for Q gives $T = \dfrac{Mmg(1+\cos\theta)}{m+M} > 0$, so the string really is taut throughout.
- Numerically, with $m = 0.8$, $M = 1$ and $a = 1$ (ratio $0.8 > 0.714$), stepping the equation of motion and computing R from Newton's second law matches the formula to 10^{-13}, and R stays non-negative. With $m/M = \frac{2}{3}$, R is negative near the top.

Common slips

- Forgetting that Q also has kinetic energy, so writing $\frac{1}{2}mv^2$ instead of $\frac{1}{2}(m+M)v^2$.
- Putting the tension into the radial equation. On a smooth surface the string runs along the tangent at P.
- Checking $R \geq 0$ only at $\theta = \frac{1}{2}\pi$ without justifying why that is enough. The concavity of f is what makes it enough.

Beyond STEP

Here energy conservation stood in for solving the equation of motion $(m+M)a\ddot{\theta} = g(M - m\cos\theta)$. This is the idea of a first integral, which becomes central in Lagrangian mechanics: the string is a constraint, and one coordinate θ describes the whole system. The bead-leaving-a-sphere family of problems, of which this is one, is treated carefully in Kleppner and Kolenkow, *An Introduction to Mechanics*.

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